Splunk Enterprise Portfolio

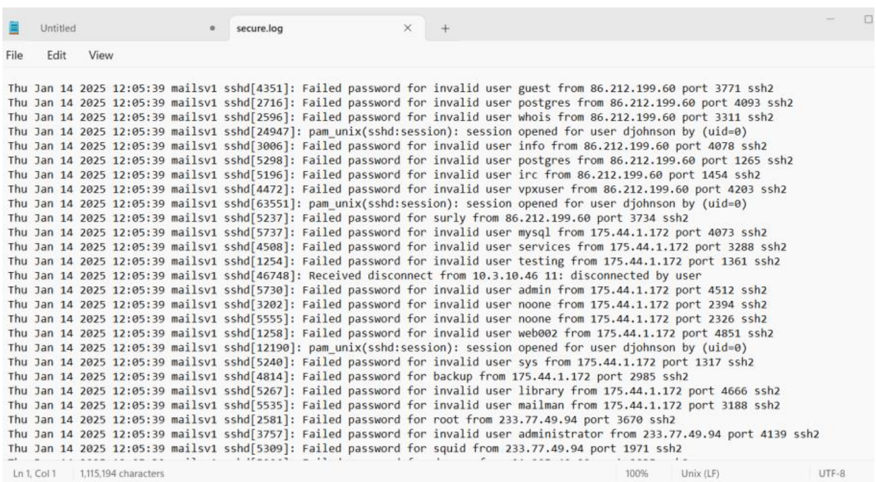
This portfolio showcases my hands-on experience with Splunk through real-world exercises.

I want to demonstrate my ability to analyze security logs, detect threats, create dashboards, and automate alerts.

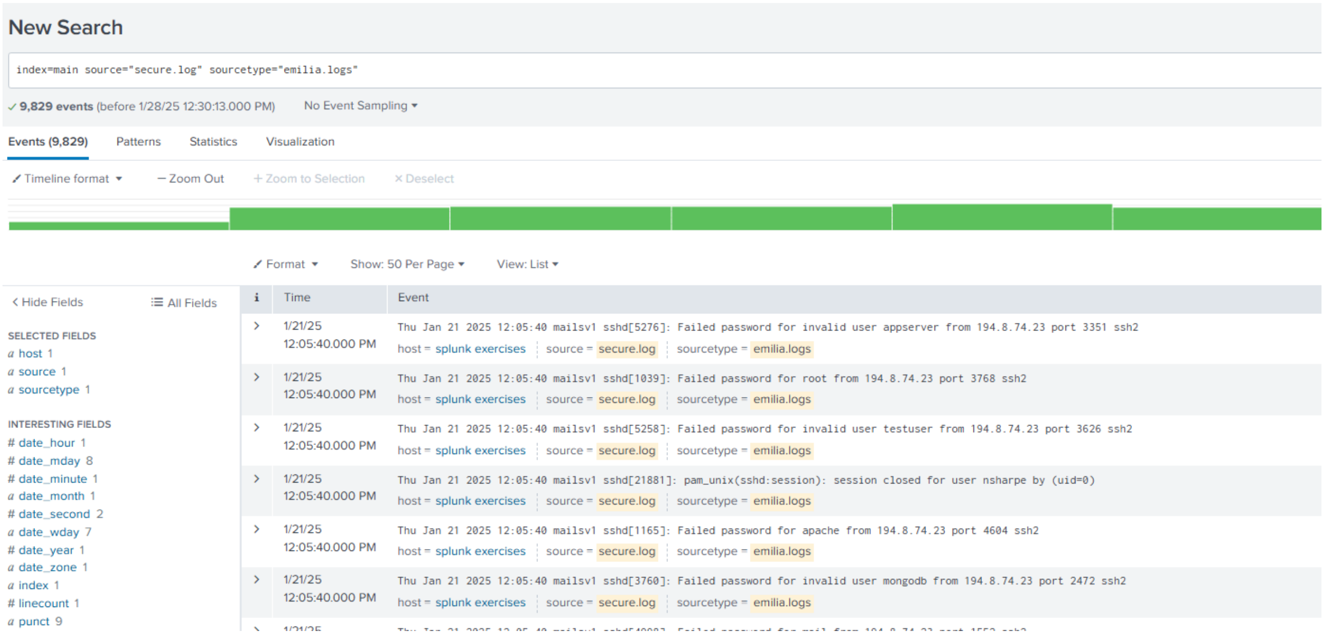
EXERCISE 1

Note: The data which I used was downloaded from here we have the following details:

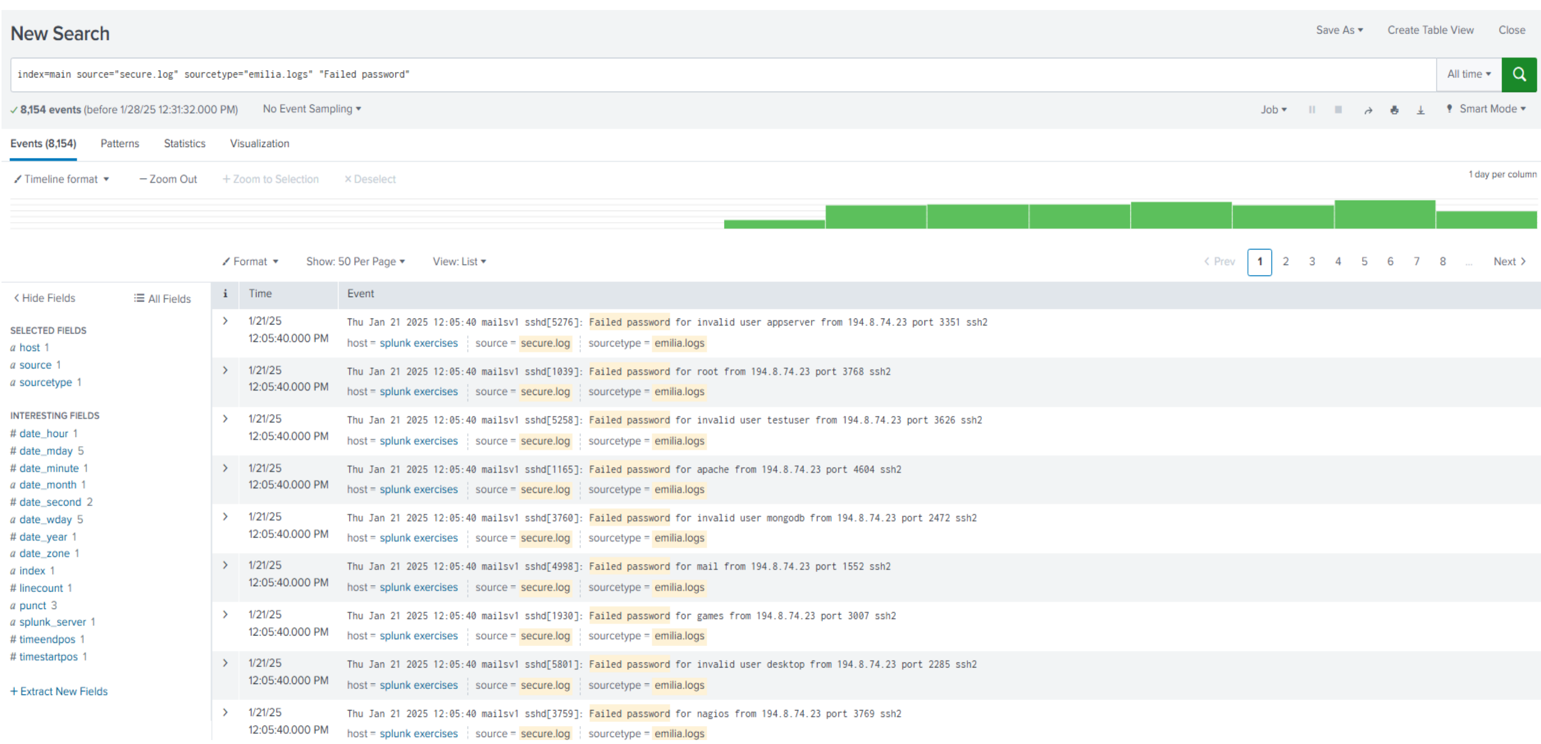
- ✓ **Source**: secure.log
- ✓ **Sourcetype**: emilia.logs
- ✓ **Host**: splunk exercises



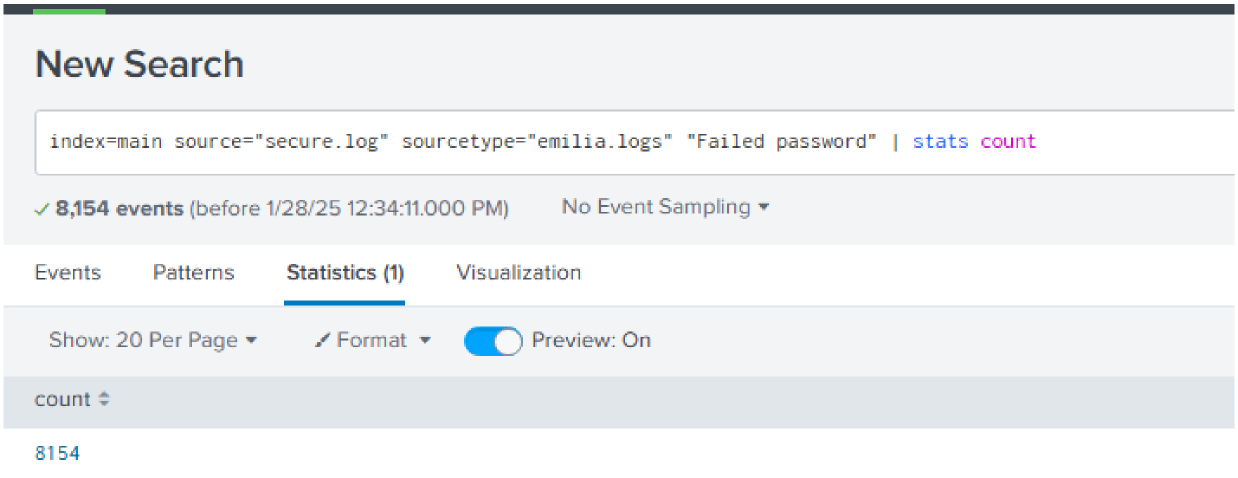

```
Thu Jan 14 2025 12:05:39 mailsv1 sshd[4351]: Failed password for invalid user guest from 86.212.199.60 port 3771 ssh2
Thu Jan 14 2025 12:05:39 mailsv1 sshd[2716]: Failed password for invalid user postgres from 86.212.199.60 port 4093 ssh2
Thu Jan 14 2025 12:05:39 mailsv1 sshd[2596]: Failed password for invalid user whois from 86.212.199.60 port 3311 ssh2
Thu Jan 14 2025 12:05:39 mailsv1 sshd[24947]: pam_unix(sshd:session): session opened for user djohnson by (uid=0)
Thu Jan 14 2025 12:05:39 mailsv1 sshd[3006]: Failed password for invalid user info from 86.212.199.60 port 4078 ssh2
Thu Jan 14 2025 12:05:39 mailsv1 sshd[5298]: Failed password for invalid user postgres from 86.212.199.60 port 1265 ssh2
Thu Jan 14 2025 12:05:39 mailsv1 sshd[5196]: Failed password for invalid user irc from 86.212.199.60 port 1454 ssh2
Thu Jan 14 2025 12:05:39 mailsv1 sshd[4472]: Failed password for invalid user vpxuser from 86.212.199.60 port 4203 ssh2
Thu Jan 14 2025 12:05:39 mailsv1 sshd[63551]: pam_unix(sshd:session): session opened for user djohnson by (uid=0)
Thu Jan 14 2025 12:05:39 mailsv1 sshd[5237]: Failed password for surly from 86.212.199.60 port 3734 ssh2
Thu Jan 14 2025 12:05:39 mailsv1 sshd[5737]: Failed password for invalid user mysql from 175.44.1.172 port 4073 ssh2
Thu Jan 14 2025 12:05:39 mailsv1 sshd[4508]: Failed password for invalid user services from 175.44.1.172 port 3288 ssh2
Thu Jan 14 2025 12:05:39 mailsv1 sshd[1254]: Failed password for invalid user testing from 175.44.1.172 port 1361 ssh2
Thu Jan 14 2025 12:05:39 mailsv1 sshd[46748]: Received disconnect from 10.3.10.46 11: disconnected by user
Thu Jan 14 2025 12:05:39 mailsv1 sshd[5730]: Failed password for invalid user admin from 175.44.1.172 port 4512 ssh2
Thu Jan 14 2025 12:05:39 mailsv1 sshd[3202]: Failed password for invalid user noone from 175.44.1.172 port 2394 ssh2
Thu Jan 14 2025 12:05:39 mailsv1 sshd[5555]: Failed password for invalid user noone from 175.44.1.172 port 2326 ssh2
Thu Jan 14 2025 12:05:39 mailsv1 sshd[1258]: Failed password for invalid user web002 from 175.44.1.172 port 4851 ssh2
Thu Jan 14 2025 12:05:39 mailsv1 sshd[12190]: pam_unix(sshd:session): session opened for user djohnson by (uid=0)
Thu Jan 14 2025 12:05:39 mailsv1 sshd[5240]: Failed password for invalid user sys from 175.44.1.172 port 1317 ssh2
Thu Jan 14 2025 12:05:39 mailsv1 sshd[4814]: Failed password for backup from 175.44.1.172 port 2985 ssh2
Thu Jan 14 2025 12:05:39 mailsv1 sshd[5267]: Failed password for invalid user library from 175.44.1.172 port 4666 ssh2
Thu Jan 14 2025 12:05:39 mailsv1 sshd[5535]: Failed password for invalid user mailman from 175.44.1.172 port 3188 ssh2
Thu Jan 14 2025 12:05:39 mailsv1 sshd[2581]: Failed password for root from 233.77.49.94 port 3670 ssh2
Thu Jan 14 2025 12:05:39 mailsv1 sshd[3757]: Failed password for invalid user administrator from 233.77.49.94 port 4139 ssh2
Thu Jan 14 2025 12:05:39 mailsv1 sshd[5309]: Failed password for squid from 233.77.49.94 port 1971 ssh2
```



1) Finding failed logins:

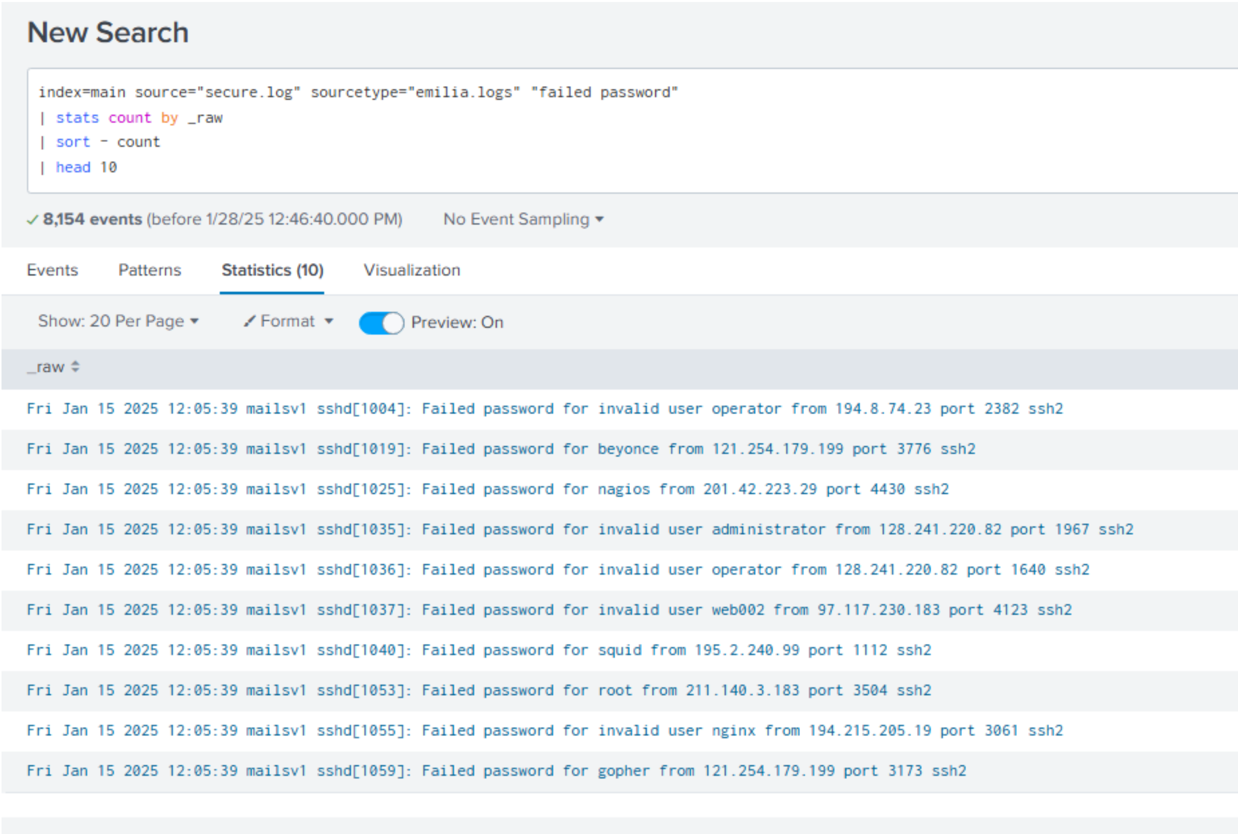


It came out that we have about 8154 failed passwords, which is a lot!



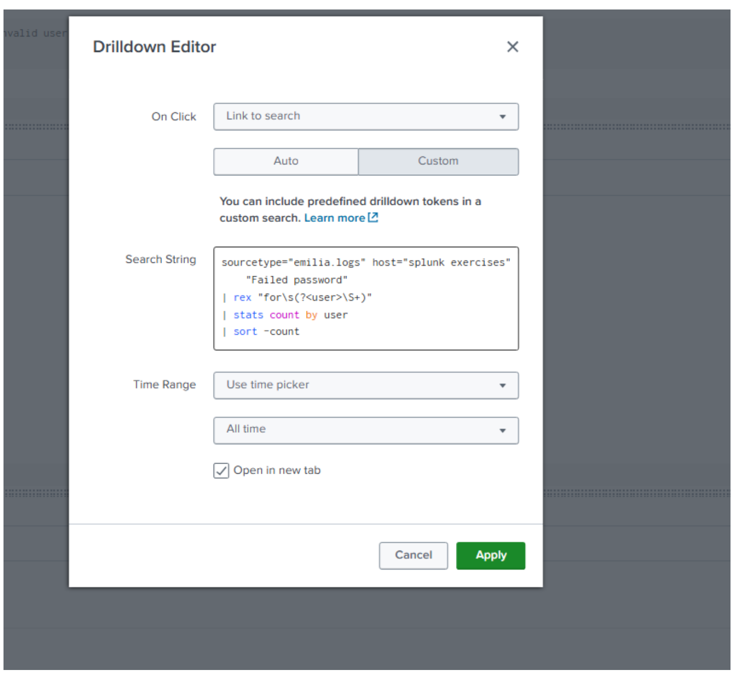
2) Finding the 10 most frequent failed password attempts

- Initiated in order to help us identifying patterns or anomalies in the log
- The conclusion would be that several attempts are listed under users like *operator, administrator, nginx, web002*, which make us think that these could be brute-force attacks, where attackers try many different user accounts.
- On the other hand, the facts that all the failed attempts are happening at the same time, could indicate that an automated tool was being used to guess passwords.
- Also, the failed attempts were caused by a variety of source IP addresses, suggesting that these are likely remote access attempts (SSH).

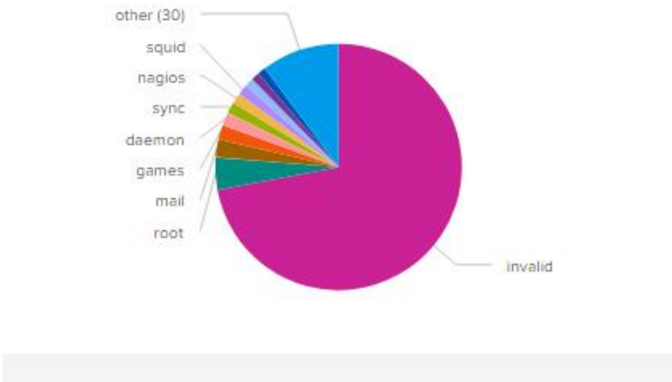


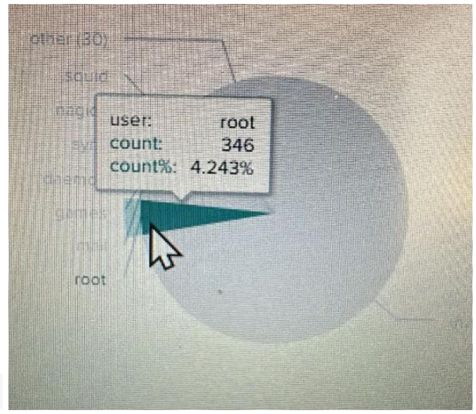
3) Creating the dashboard with the top users with failed logins

I`ve used the following query



And I`ve obtained the following chart

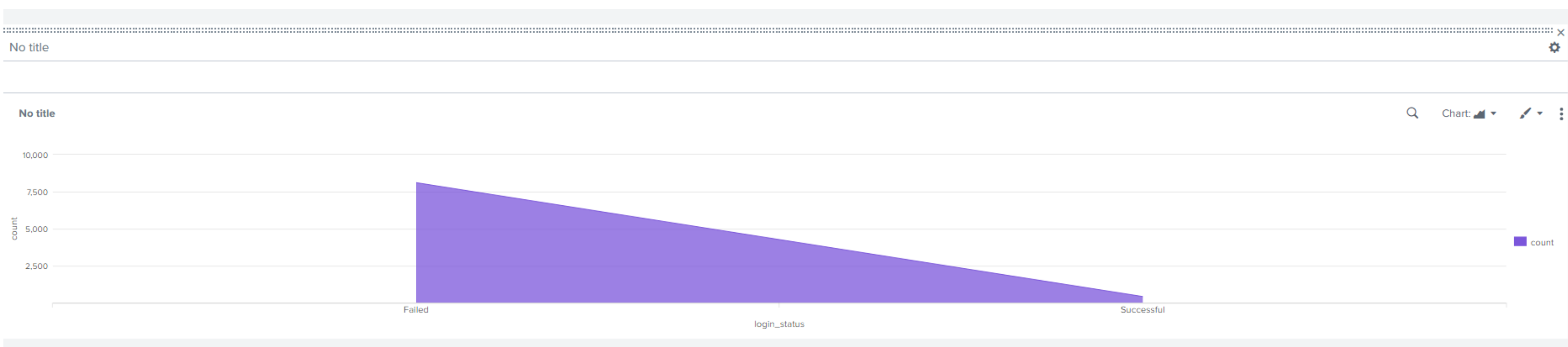


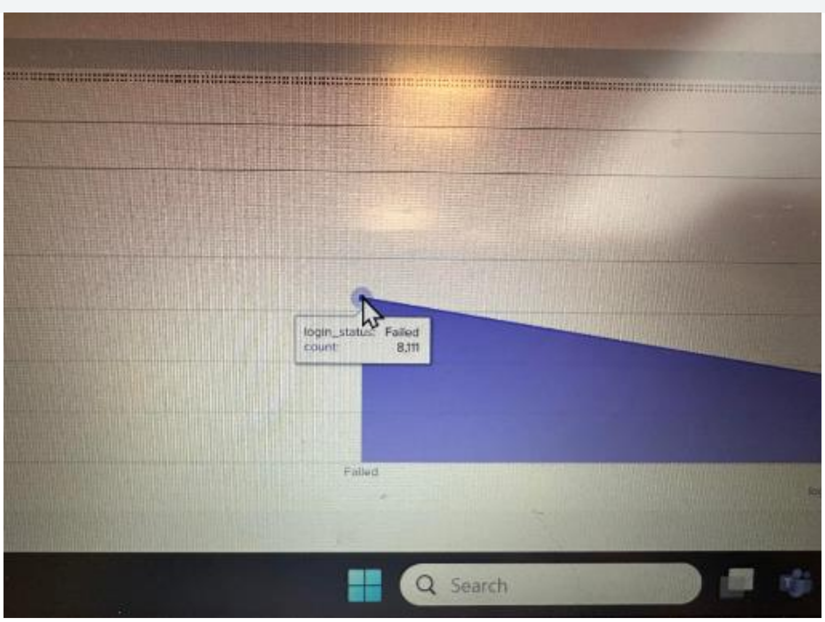


4) Failed versus successful logins

I`ve used the following string in order to view on the chart the difference between failed and successful logins.

```
sourcetype="emilia.logs" host="splunk exercises" ("Failed password" OR "Accepted password")
| eval login_status=if(match(_raw, "Failed password"), "Failed", "Successful")
| stats count by login_status
```





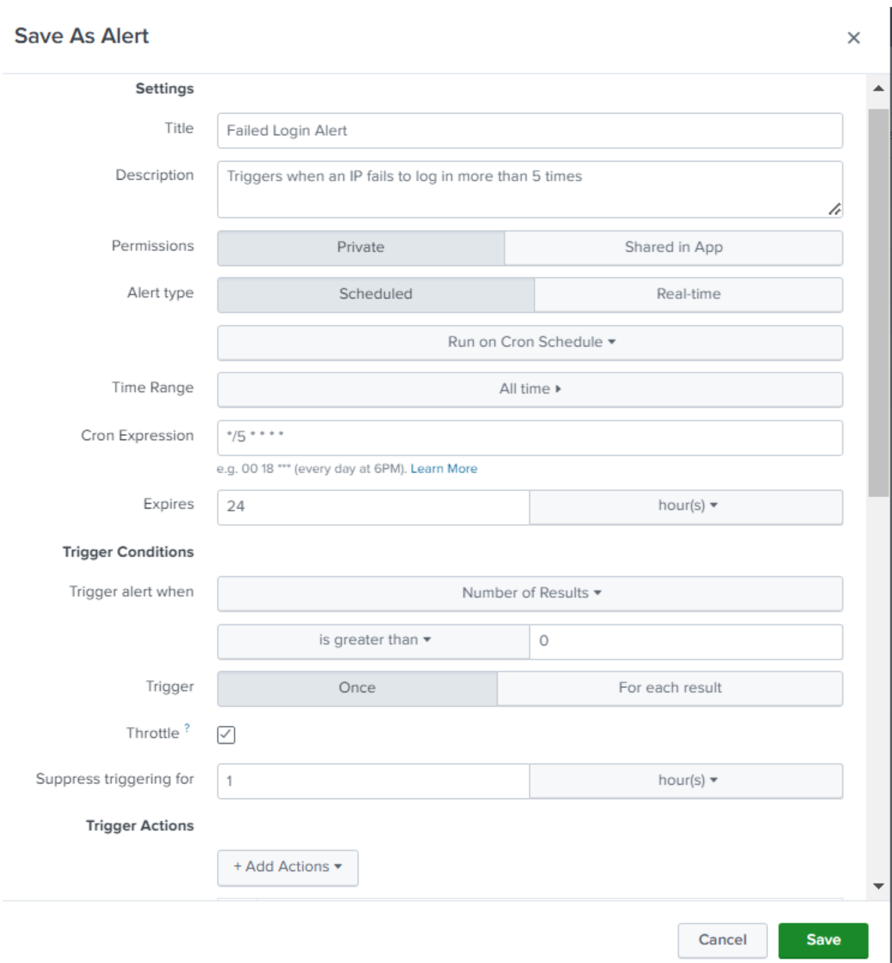
The number of failed logins was 8.111 while the successful logins were 423. (*as per the report created and used)

Advices:

I would recommend to review who has access to SSH and to harden the authentication methods; instead of passwords we could use SSH keys or other MFA.

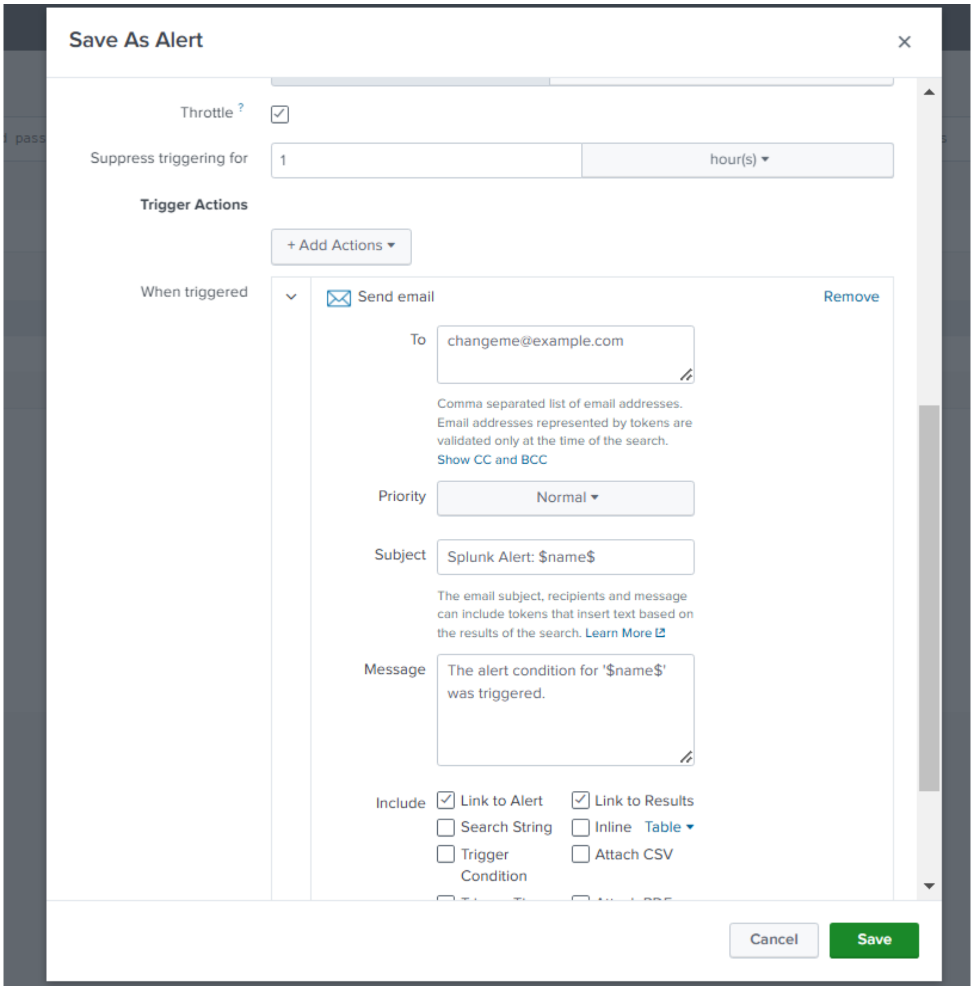
The other advice would be to block certain suspicious IP adresses, which are blocked or flagged in firewall logs.

And last but not the least, to set up alerts in Splunk for repeated failed login attempts, thing which I already did (as shown below).



Since we suspect a brute force attack, setting the interval to every 5 minutes could be a good option.

Also, I checked the option "Throttle" in order to mute alerts for the same IP for 1 hour to prevent alert spam.


Save As Alert
Throttle
Suppress triggering for
1
hour(s)
Trigger Actions
+ Add Actions
When triggered
Send email
Remove
To
changeme@example.com
Comma separated list of email addresses. Email addresses represented by tokens are validated only at the time of the search.
Show CC and BCC
Priority
Normal
Subject
Splunk Alert: $name$
The email subject, recipients and message can include tokens that insert text based on the results of the search. Learn More
Message
The alert condition for '$name$' was triggered.
Include
Link to Alert
Link to Results
Search String
Inline
Table
Trigger Condition
Attach CSV
Cancel
Save

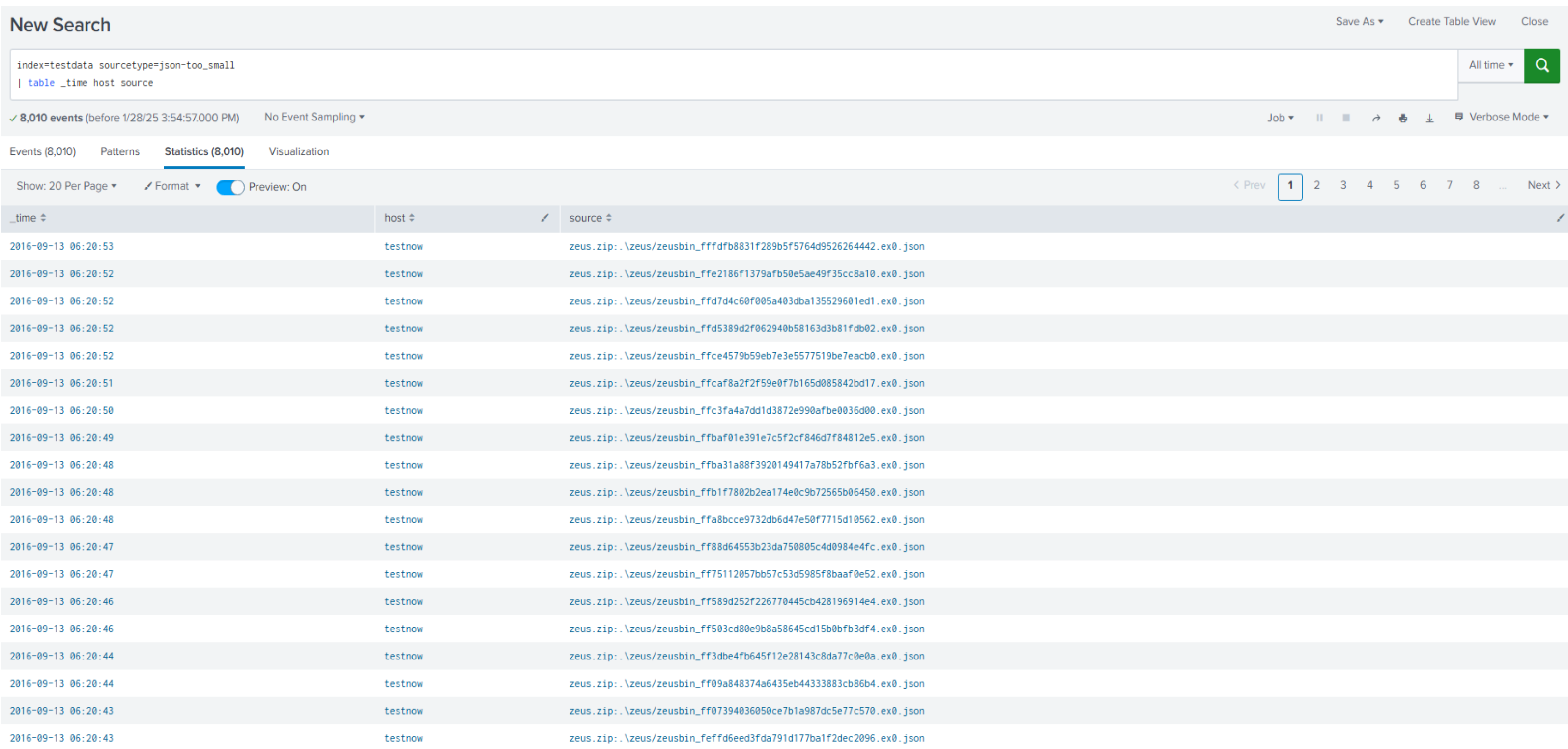
EXERCISE 2

Note: The data used for analysis was taken from here *(Static information about Zeus binaries)* and uploaded in Splunk under these details:

- ✓ **Source/file name**: zeus.zip
- ✓ **Sourcetype**: json-too_small
- ✓ **Host**: testnow

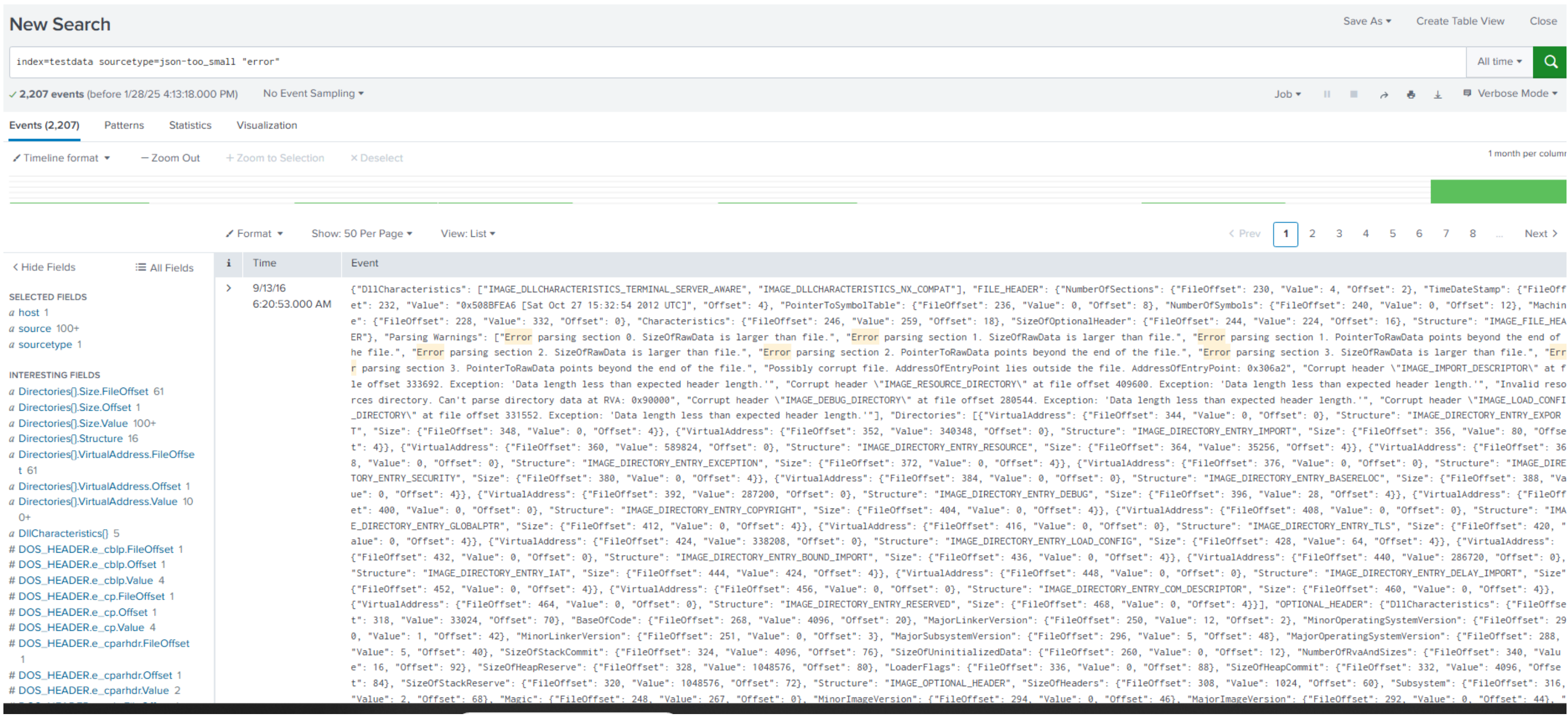
We will perform a malware analysis.

1) First we will check when the events occurred and where they came from; we`ve noticed that they all have the same host.



2) I`ve found no “attack”, hence 2207 events with “error”.

The "error parsing" could indicate possible file corruption; one more indication of this would be *IMAGE_SCN_MEM_WRITE* and *IMAGE_SCN_MEM_EXECUTE*, which could lead to obfuscation.



<u>Recommendations</u>:

- to implement File Integrity Monitoring, to monitor system files or unexpected file access patterns

-to implement EDR against suspicious memory-based activity due to IMAGE_SCN_MEM_WRITE and IMAGE_SCN_MEM_EXECUTE flags

- to use if it`s the case sandboxing tools for testing suspicious files or activities

- to use application whitelisting so that only trusted, pre-approved applications are allowed to run